

PRISM

AI-Powered Food Safety Early-Warning Platform

Complete Ideathon Master Guide

One document for the entire team: problem, users, solution, features, AI, food inspector workflow, Evidence Vault, risk engine, clustering, hotspot detection, architecture, database, MVP, implementation plan, demo, business model, safety, and judge questions.

Core idea: PRISM connects citizen reports, AI-assisted analysis, evidence integrity, risk scoring, complaint clustering and food-inspector review into one continuous loop.

Team rule: Build one convincing end-to-end journey instead of many disconnected features.

1. Executive Summary

PRISM is a citizen-powered food-safety early-warning platform. A citizen can report a possible food-safety issue using a photo and simple description. PRISM uses AI to analyze visible signals and structure the complaint, stores supporting evidence in an integrity-aware evidence record, calculates risk signals, groups similar complaints, and highlights potential hotspots to a food inspector or authorized reviewer.

The inspector then reviews the evidence, investigates the case, adds inspection evidence, records findings and updates the case status. PRISM therefore creates a feedback loop from citizen → intelligence → human investigation → citizen status.

Important positioning: PRISM is a decision-support and reporting platform. AI does not prove contamination, declare a business guilty, or replace official food-safety procedures.

2. The Problem

Food-safety problems are often discovered after a customer notices an issue, complains, or becomes ill. Ordinary citizens may find reporting difficult, may not know which information is useful, and may have little visibility into what happens after submission.

At the same time, individual complaints can remain fragmented. A single report may look minor, while many similar reports from the same vendor or area can indicate an emerging pattern.

Core problem statement: How can we make food-safety reporting simple for ordinary citizens and convert scattered reports into early-warning signals that help authorized food-safety teams prioritize investigation sooner?

3. Why Existing Processes Leave a Gap

The technical blueprint supplied with this project describes three commonly separate tracks: a citizen complaint portal, an evidence model that may simply attach files to a ticket, and backend risk-scoring systems that can operate separately from the citizen reporting experience.

Its central observation is that PRISM attempts to fuse these tracks into one loop: the same citizen action can open a case, contribute to evidence, update risk signals, and feed complaint clustering and hotspot detection.

PRISM's opportunity: connect the citizen's real-world narrative and evidence with the intelligence layer used to prioritize human review.

4. The PRISM Solution

One-line pitch: “PRISM turns citizen food-safety reports into early-warning signals by combining AI-assisted reporting, structured evidence, dynamic risk scoring, complaint clustering, hotspot detection and human inspector review.”

PRISM is designed as three connected experiences:

Experience	Purpose
■ Citizen	Report a possible issue easily, provide evidence, and track status.
■ PRISM AI	Analyze visible evidence, structure the report, cluster similar reports, and produce risk signals.
■ Food Inspector	Prioritize cases, review evidence, investigate, add findings, and update status.

5. Complete End-to-End Architecture

Citizen → Report → AI Analysis → Complaint Assistant → Evidence Vault → Risk Engine → Complaint Clustering → Hotspot Detection → Inspector Dashboard → Human Investigation → Status Feedback

A simple mental model for the team:

■ Evidence → ■ Understanding → ■ Pattern → ■ Priority → ■ Investigation → ■ Feedback

6. User 1 — Citizen

The citizen is the primary MVP user. The interface should be fast enough that someone can create a useful report without understanding food-safety terminology.

Home:

- Report Food Issue
- Check Food / Product (optional)
- My Reports

Report flow:

- Take/upload photo or video
- Describe the issue in normal language
- Confirm restaurant/vendor
- Confirm location
- Confirm food/product
- Provide incident date/time
- Upload bill/packaging if available
- Review and submit

Citizen promise: PRISM should ask for only the information that is still missing.

7. AI Complaint Assistant

The blueprint recommends a guided slot-filling agent, not a free-roaming chatbot. Its job is to fill six required fields and then hand a clean structured object to the Review & Submit screen.

Required field	Possible source
vendor_name	User / OCR of bill
location	User / device GPS
food_item	User / image analysis
issue_type	Image analysis first; user confirms
incident_datetime	User
evidence_available	User: photo / bill / both / none

Conversation rule: Run image analysis first. If the image already provides an answer, do not ask the citizen for it again. Ask only one or two short questions at a time.

Example: User says, “I found plastic in my biryani.” PRISM asks where it was purchased and whether a bill is available, rather than asking the user to complete a long form.

8. AI Image Analysis

The vision component should look only for what is visibly present in the uploaded image. Suitable prototype categories include foreign objects, mold-like growth, pest evidence, spoilage/discoloration and packaging damage.

Output	Example
detected_issue	possible_foreign_object
risk_level	High
confidence	0.91
explanation	Visible object appears inconsistent with the food item.

Critical limitation: A photo cannot prove laboratory contamination. The UI must use language such as “possible,” “suspected,” or “reported.”

9. Evidence Vault

The Evidence Vault is more than an attachments folder. The supplied blueprint recommends hashing every uploaded file on server-side ingestion, keeping evidence records immutable, and maintaining an append-only audit log.

Ingestion flow:

Upload → SHA-256 hash → metadata → object storage → evidence record → audit event

Evidence data	Purpose
SHA-256 hash	Detect later changes to file bytes
Report ID	Links evidence to the case
Uploader	Records who supplied it
GPS	Records available location context
Upload timestamp	Records server-side ingestion time
Capture timestamp	Optional EXIF timestamp when available
Audit log	Records create/view/export/download events

Access model: Citizens can view but not alter/delete their evidence through the application; authorized reviewers can view evidence associated with cases in their jurisdiction. The prototype should not claim to be a complete legal-grade digital evidence system.

10. Dynamic Risk Engine

PRISM should not declare a restaurant unsafe because of one unverified complaint. Instead, it calculates a triage signal from multiple factors.

Signal	What it contributes
Complaint frequency	How many reports are associated with a vendor or area
Similarity	How many reports describe a related issue
Recent increase	Whether complaints are rising unusually fast
Severity	Whether the reported issue warrants higher priority
Verification status	Whether reports have been reviewed/verified

Signal	What it contributes
Location pattern	Whether reports are concentrated geographically

Example: ABC Restaurant → 14 reports → 9 similar → 8 in the last 7 days → +180% increase → risk score 87/100 → HIGH PRIORITY.

The score is a prioritization signal for human review, not a legal or scientific determination.

11. Complaint Clustering

PRISM groups reports that may describe the same or related issue. The prototype can use semantic text embeddings plus additional metadata such as vendor, time and location.

Example:

- “Bad smell from chicken”
- “Chicken tasted spoiled”
- “Food had a strange smell”
- “Felt unwell after eating there”

If other signals align, the system can create a possible incident cluster instead of treating the reports as unrelated tickets.

Output: “Possible incident cluster — 14 related reports.”

12. Hotspot Detection & Map

The map converts report activity into geographic signals. For an ideathon, synthetic/demo data can be used, but it must be clearly labeled as DEMO DATA.

Map element	Meaning
Green	Lower observed report activity
Yellow	Moderate activity
Red	Potential emerging hotspot

Clicking a hotspot can show report count, establishments involved, issue categories, recent growth, clusters and priority cases.

Important: A red hotspot is a signal for investigation, not proof that every business or person in that area is unsafe.

13. User 2 — Food Inspector

The food inspector is a major user of PRISM. The inspector receives the intelligence generated by the citizen and AI layers, but retains human responsibility for investigation and decisions.

Inspector flow:

Priority queue → Case details → Evidence → Clusters → Hotspot → Investigation → Additional evidence → Findings → Status update → Citizen feedback

14. Inspector Dashboard

Dashboard section	Example
High priority	12
Medium priority	34

Dashboard section	Example
Low priority	87
Potential hotspots	5
New clusters	3

The dashboard should answer three questions immediately:

- What needs attention first?
- Where are possible emerging hotspots?
- Why did PRISM prioritize this case?

15. Inspector Case View

Section	Information
Business	Restaurant/shop/brand, location
Reports	Individual and related reports
Evidence	Photos, bills, packaging and audit information
AI assessment	Possible issue, confidence, explanation
Risk	Score and contributing factors
Cluster	Similar reports and pattern details
Investigation	Inspector notes and findings
Status	New, under review, investigation, verified, not verified, actioned, closed

16. Inspector Investigation Mode

- Visit establishment
- Verify relevant licence/details according to applicable process
- Inspect kitchen hygiene
- Check storage conditions
- Check food handling
- Record observations
- Upload appropriate inspection evidence
- Record findings and next step

The exact regulatory procedure depends on the competent authority. PRISM should support the workflow rather than invent regulatory powers.

17. Inspector Evidence Upload

The inspector can add new evidence such as inspection photographs, documents and notes. New evidence should be created as a new immutable record rather than modifying an existing evidence row.

The same evidence-integrity principles apply: SHA-256 hash, uploader, timestamp, location where available, and audit events.

Why this matters: the evidence chain continues from citizen submission to inspector investigation instead of splitting the case into unrelated systems.

18. Human Verification & Decision

AI signal → Inspector review → Investigation → Human decision

PRISM can recommend that a case deserves attention. It must not automatically declare contamination, guilt, or enforcement action.

If a citizen describes illness symptoms, the assistant should encourage appropriate medical attention and continue collecting the report separately; it must not diagnose the user.

19. Citizen Feedback Loop

After the authorized reviewer updates the case, the citizen sees a simple status timeline.

Report #PR-10482

- ✓ Submitted
- ✓ Evidence received
- ✓ Under investigation

Current status: UNDER INVESTIGATION

This feedback loop is important because the citizen is not just a data source; they are a user who should understand what happened to their report.

20. Optional Restaurant / Food Business Module

This should be a secondary or future module, not the core ideathon build.

- Complaint trends
- Repeated issue categories
- Risk indicators
- Corrective-action checklist
- Improvement tracking
- Multi-location analytics

This module creates a possible B2B SaaS path while keeping the citizen and inspector workflows central.

21. Optional Product / Packaged Food Check

A consumer can scan a barcode or package and view available product information. Reliable official sources should be used for recalls or alerts in a production system. Demo data can be used during the ideathon if clearly labeled.

This gives PRISM a recurring consumer utility beyond complaint submission.

22. Complete Technical Architecture

Layer	Suggested technology
Frontend	Next.js + Tailwind CSS, or React + Tailwind
Backend	Python FastAPI
Database	Supabase / PostgreSQL
AI	Multimodal vision + language model

Layer	Suggested technology
Embeddings / clustering	Sentence-transformers + clustering algorithm for prototype
Maps	Mapbox or Google Maps
Storage	Supabase Storage or S3-compatible object storage
Authentication	Supabase Auth
Hosting	Vercel for frontend plus suitable backend hosting

Do not choose technologies just to make the stack look impressive. The goal is a reliable end-to-end prototype.

23. AI Pipeline

1. Input: photo + text + optional bill/product information
2. Extraction: identify vendor, food, issue and key facts
3. Vision: analyze visible signals
4. Evidence check: identify available/missing evidence
5. Risk features: frequency, similarity, recency, severity, location and verification
6. Clustering: group related reports
7. Risk scoring: produce triage priority
8. Dashboard: surface cases, clusters and hotspots

24. Suggested Database

Table	Core fields
users	id, name, contact, role
reports	id, user_id, vendor_id, issue_type, description, date, location, status
vendors	id, name, address, latitude, longitude
evidence	id, report_id, file_url, file_type, sha256_hash, uploaded_by, geo_lat, geo_lng, upload_timestamp
evidence_audit_log	id, evidence_id, action, actor, timestamp
ai_assessments	report_id, detected_issue, risk_level, confidence, explanation
clusters	id, criteria, report_count, created_at
risk_scores	vendor/area, score, factors, updated_at
investigations	case_id, inspector, notes, findings, status, timestamps

25. Evidence Vault Technical Rules

- Hash every uploaded file with SHA-256 server-side before storage processing.
- Do not edit an evidence row after creation; corrections create a new row.
- Use object storage with versioning where possible.
- Maintain an append-only audit log for CREATE, VIEW, EXPORT and DOWNLOAD.
- Store uploader, GPS and upload timestamp.
- Retain capture timestamp from EXIF when available.
- Restrict application-layer deletion rights.
- Use access control based on user role and case jurisdiction.

26. Complaint Assistant Technical Rules

- Use guided slot filling, not open-ended autonomous conversation.
- Run image analysis before asking questions.
- Never re-ask fields already known.
- Ask at most one or two short questions per turn.
- Use possible/suspected/reported language.
- Never diagnose illness.
- When all fields are filled, emit structured report data for Review & Submit.
- Keep the final citizen confirmation step explicit.

27. Risk Engine — Prototype Formula

A simple prototype can normalize several signals to 0–100 and combine them with transparent weights. The exact weights are a prototype choice and should be tuned with testing; they are not a regulatory standard.

Component	Example weight
Recent complaint volume	25%
Similarity / cluster strength	20%
Recent growth rate	20%
Reported severity	15%
Geographic concentration	10%
Verification/history signal	10%

Example: score = weighted sum of normalized signals. Always show the contributing factors so the inspector can understand why a case is high priority.

28. MVP — What We Actually Build

Feature	Priority
Citizen reporting	MUST
Photo upload	MUST
AI image/text assessment	MUST
Guided complaint assistant	MUST
Evidence Vault	MUST
Risk score	MUST
Complaint clustering	MUST
Hotspot map	MUST
Inspector dashboard	MUST
Inspector investigation/status	MUST
Restaurant dashboard	FUTURE
Predictive analytics	FUTURE

Feature	Priority
Product check	OPTIONAL

29. Implementation Plan

Phase	Build
1 — Foundation	Frontend structure, authentication, database schema
2 — Citizen reporting	Report form, photo/evidence upload, location, My Reports
3 — AI intake	Vision analysis + guided complaint assistant
4 — Evidence Vault	SHA-256 hashing, storage, metadata, audit log
5 — Intelligence	Risk score + complaint clustering
6 — Inspector	Priority queue, case view, evidence, investigation, status
7 — Map	Hotspot aggregation and map visualization
8 — Integration	Connect citizen → AI → evidence → risk → inspector → status
9 — Polish	UI, testing, error handling, demo data, pitch

Development principle: get one complete case working before adding advanced features.

30. Team Division

Role	Responsibility
Frontend developer	Citizen app, report screens, inspector UI
Backend developer	APIs, database, authentication, report lifecycle
AI developer	Vision analysis, structured extraction, assistant
Data/Intelligence	Risk score, clustering, hotspot logic
Maps/UI/Pitch	Map, charts, design consistency, presentation and demo

For smaller teams, combine Frontend + Design and Backend + Data.

31. Demo Data Strategy

For the ideathon, create a small synthetic dataset containing multiple restaurants, dates, issue types, coordinates, and reports. Clearly display “DEMO DATA” in the dashboard.

Example dataset story:

- ABC Restaurant has 14 reports.
- 9 reports are semantically similar.
- 8 arrived within 7 days.
- Complaint activity has increased rapidly.
- Several reports contain photos/bills.
- PRISM creates a high-priority cluster and hotspot.

The purpose is to demonstrate the system's logic, not to make unsupported claims about a real business.

32. 3-Minute Winning Demo

0:00–0:30 — Citizen: “I found a foreign object in my biryani.” Upload image.

0:30–1:00 — AI: Show possible issue, confidence and priority. Assistant asks only missing questions.

1:00–1:20 — Evidence: Show report, bill/photo, timestamp and SHA-256 hash.

1:20–1:50 — Intelligence: Show risk score and matching complaint cluster.

1:50–2:20 — Inspector: Open high-priority case, evidence and hotspot map.

2:20–2:45 — Investigation: Add an inspector observation/photo and update status.

2:45–3:00 — Citizen: Show status update. Finish with the one-line pitch.

33. Exact Demo Story

Citizen: “I ordered chicken biryani from ABC Restaurant and found something unusual.”

PRISM AI: “Possible foreign object detected. Risk: High. Confidence: 91%.”

Assistant: “Where did you purchase it?” → “Do you have the bill?”

Evidence Vault: Evidence ID, SHA-256, timestamp, location, report link.

Risk Engine: “87/100 — High priority.”

Cluster: “14 related reports; 9 similar.”

Inspector: “Review case → Start investigation → Add evidence → Under Investigation.”

Citizen: “My report is now under investigation.”

34. What Makes PRISM Different?

Typical complaint experience	PRISM
Individual ticket	Connected case + intelligence signal
Manual form-heavy intake	AI-assisted structured intake
Photo as attachment	Integrity-aware evidence record
Complaints viewed separately	Semantic complaint clustering
Backend score hidden from citizen	Explainable prioritization signals
Risk and complaint systems disconnected	One integrated loop
Authority sees tickets	Inspector sees priority, evidence, clusters and hotspots

The strongest differentiator is not simply “AI.” It is the **integration of citizen reporting, evidence, pattern detection and human inspection in one workflow.**

35. Startup Potential

PRISM can evolve into a food-safety intelligence platform.

Customer	Possible offering
Consumers	Free reporting, evidence and safety tools
Restaurants	Safety analytics and corrective-action dashboard
Food chains	Multi-location monitoring and trend analytics

Customer	Possible offering
Institutions / authorities	Appropriate analytics and workflow partnerships

Do not promise government adoption or automatic enforcement. Present those as future partnership/integration opportunities.

36. Trust, Privacy & Safety

- AI assessments are not laboratory confirmation.
- Do not publicly label businesses unsafe from unverified reports.
- Use neutral language: reported, possible, suspected, requires review.
- Minimize collection of personal data.
- Use role-based access controls.
- Clearly distinguish demo/synthetic data from real-world data.
- Do not fabricate official government submissions, inspections or outcomes.
- Keep citizen evidence and inspector evidence auditable.

37. Judge Questions & Strong Answers

Q: Why can't users simply use existing food-safety complaint channels?

A: They can. PRISM is not trying to replace official channels. It reduces reporting friction, structures evidence and adds pattern/risk intelligence so multiple reports can become useful early-warning signals.

Q: What is innovative?

A: The integrated loop: AI-assisted intake, evidence integrity, semantic complaint clustering, dynamic risk signals, hotspot detection and human inspector review in one system.

Q: Can AI prove contamination from an image?

A: No. PRISM only identifies possible visible issues. Confirmation requires appropriate human investigation and, where applicable, laboratory or official processes.

Q: Does AI decide whether a restaurant is guilty?

A: No. AI produces signals and prioritization. The food inspector or authorized authority performs human review and decides the appropriate action.

Q: Why is the Evidence Vault needed?

A: It makes evidence more structured and auditable than a simple attachment. Hashes help detect file changes, and audit logs record access events.

Q: Why does the inspector need PRISM?

A: It brings reports, evidence, clusters, risk factors and hotspots into one prioritized workspace.

Q: What if the AI is wrong?

A: The result is explicitly treated as a possible signal. The inspector can review and reject it. Original evidence remains available.

Q: Where does the data come from?

A: The ideathon prototype can use clearly labeled synthetic/demo data. A production deployment would need appropriate official integrations, data-sharing arrangements and privacy controls.

Q: How do you monetize?

A: Keep core citizen reporting free and offer paid analytics/monitoring to restaurants, food chains and appropriate institutional customers.

Q: Why should people use it if official complaint systems already exist?

A: PRISM can make the reporting experience easier while adding a layer that existing complaint tickets generally do not provide: cross-report pattern detection and risk prioritization.

38. What NOT to Build

- Food delivery or ordering
- Payments
- Social media
- A large government ERP
- Blockchain without a real requirement
- Claims of laboratory-level AI contamination detection
- Automatic punishment/enforcement
- Twenty different dashboards
- Real government integration without authorization/access

A polished, working end-to-end MVP is more valuable in an ideathon than a huge unfinished platform.

39. Final Architecture — One Picture in Words

■ CITIZEN

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■ Report: photo + text + location

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■ AI: image analysis + guided complaint assistant

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■ Evidence Vault: SHA-256 + metadata + audit log

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■ Intelligence: risk score + semantic clustering

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■■ Hotspot detection

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■ Food Inspector: priority queue + case review

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■ Investigation + additional evidence

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■ Human decision/status

↓

■ Citizen receives status update

40. Final Project Statement

PRISM is an AI-powered food-safety early-warning platform that helps citizens report potential food-safety issues, organizes and protects evidence, detects related complaint patterns, calculates risk signals, identifies emerging hotspots, and gives food inspectors a prioritized workspace for human investigation.

PRISM's philosophy is simple: **one complaint is information; a pattern of complaints can become an early warning.**

41. Final Pitch

“Today, a food-safety problem may be reported one complaint at a time. PRISM connects those small signals. A citizen takes a photo, AI helps structure the report, evidence is preserved, similar complaints are clustered, risk is assessed, and emerging hotspots are highlighted for human review. We are not replacing food-safety authorities—we are giving them an earlier and clearer signal of where to look.”

42. Final Build Checklist

- Citizen can create a report.
- Photo upload works.
- AI produces strict structured output.
- AI assistant asks only missing fields.
- Review & Submit screen works.
- Evidence receives SHA-256 hash.
- Evidence metadata is stored.
- Evidence access events are logged.
- Risk score is explainable.
- Similar complaints form a cluster.
- Hotspot appears on the map.
- Inspector can open a priority case.
- Inspector can view evidence.
- Inspector can add investigation evidence.
- Inspector can update status.
- Citizen can see status.
- Demo data is labeled DEMO DATA.
- No screen claims AI proves contamination.
- No screen claims PRISM performs official enforcement.
- The entire demo works from start to finish without manual database edits.

43. Source / Scope Note

This master guide consolidates the PRISM project direction developed in this conversation and the uploaded technical blueprint titled “FoodShield AI — Real-World Benchmark & Build Blueprint.” The uploaded blueprint covers the AI Complaint Assistant, Evidence Vault, real-world comparison, differentiation, reference implementation concepts, and a complaint-clustering bonus. The final branding in this guide is **PRISM**.

Where this guide describes future modules, prototype choices, example risk weights, demo numbers, or commercialization, those are presented as implementation recommendations rather than claims established by the uploaded source.